Using theory and experiments to understand plant physiological responses to nutrient availability

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New Phytologist

Tansley review | Open Access

Empirical evidence and theoretical understanding of ecosystem carbon and nitrogen cycle interactions

Benjamin D. Stocker, Ning Dong, Evan A. Perkowski, Pascal D. Schneider, Huiying Xu, Hugo J. de Boer, Karin T. Rebel, Nicholas G. Smith, Kevin Van Sundert, Han Wang, Sarah E. Jones ... See all authors

First published: 23 October 2024 | https://doi.org/10.1111/nph.20178 | Citations: 1

Stocker et al. (2025) *New Phyt*

Take home: we need to ground our understanding of plant-nutrient relationships in theory

Stocker et al. (2025) *New Phyt*

Talk outline

- Photosynthetic least cost theory and its relation to nutrients
- Tests of the theory
    1. The interactions between nutrient supply and demand
    2. Responses to elevated CO_2
- Conclusions

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Photosynthetic least cost theory

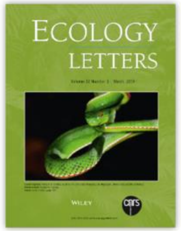
Optimally, plants will maintain fastest rate of photosynthesis at the lowest summed resource cost (water and nutrient use)

Wright et al. (2003) *Am Naturalist*

Photosynthetic least cost theory

- Can be used to predict changes in photosynthetic nitrogen in response to changes in
  1. Supply
  2. Demand

ECOLOGY LETTERS



Volume 22, Issue 3
March 2019
Pages 506-517

Letter | Open Access

Global photosynthetic capacity is optimized to the environment

Nicholas G. Smith, Trevor F. Keenan, I. Colin Prentice, Han Wang, Ian J. Wright, Ülo Niinemets, Kristine Y. Crous, Tomas F. Domingues, Rossella Guerrieri, F. Yoko Ishida, Jens Kattge ... See all authors

First published: 04 January 2019 | https://doi.org/10.1111/ele.13210 | VIEW METRICS

Global Change Biology

Volume 26, Issue 9
September 2020
Pages 5202-5216

PRIMARY RESEARCH ARTICLE | Full Access

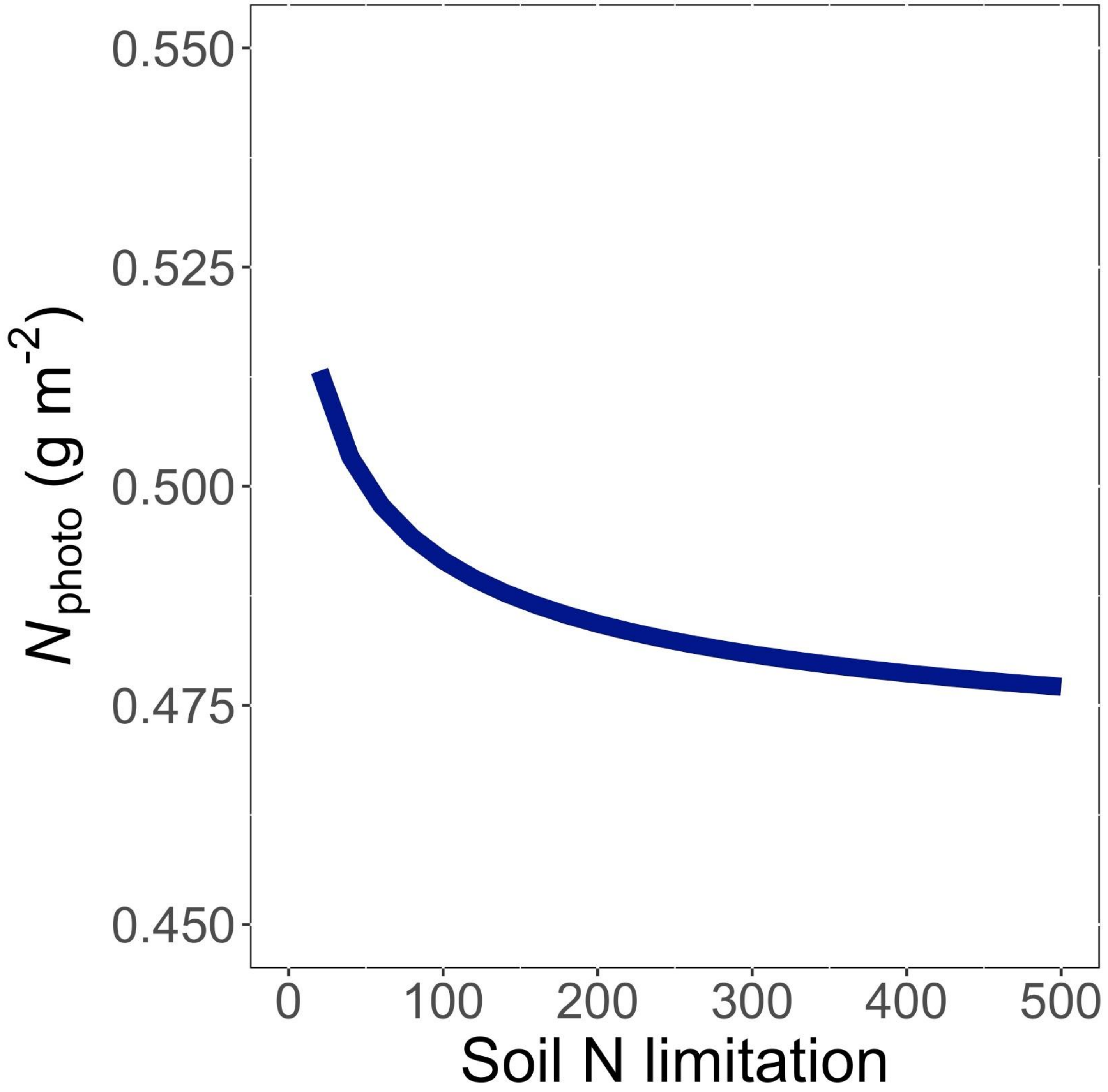
Mechanisms underlying leaf photosynthetic acclimation to warming and elevated CO_2 as inferred from least-cost optimality theory

Nicholas G. Smith, Trevor F. Keenan

First published: 11 June 2020 | https://doi.org/10.1111/gcb.15212 | VIEW METRICS

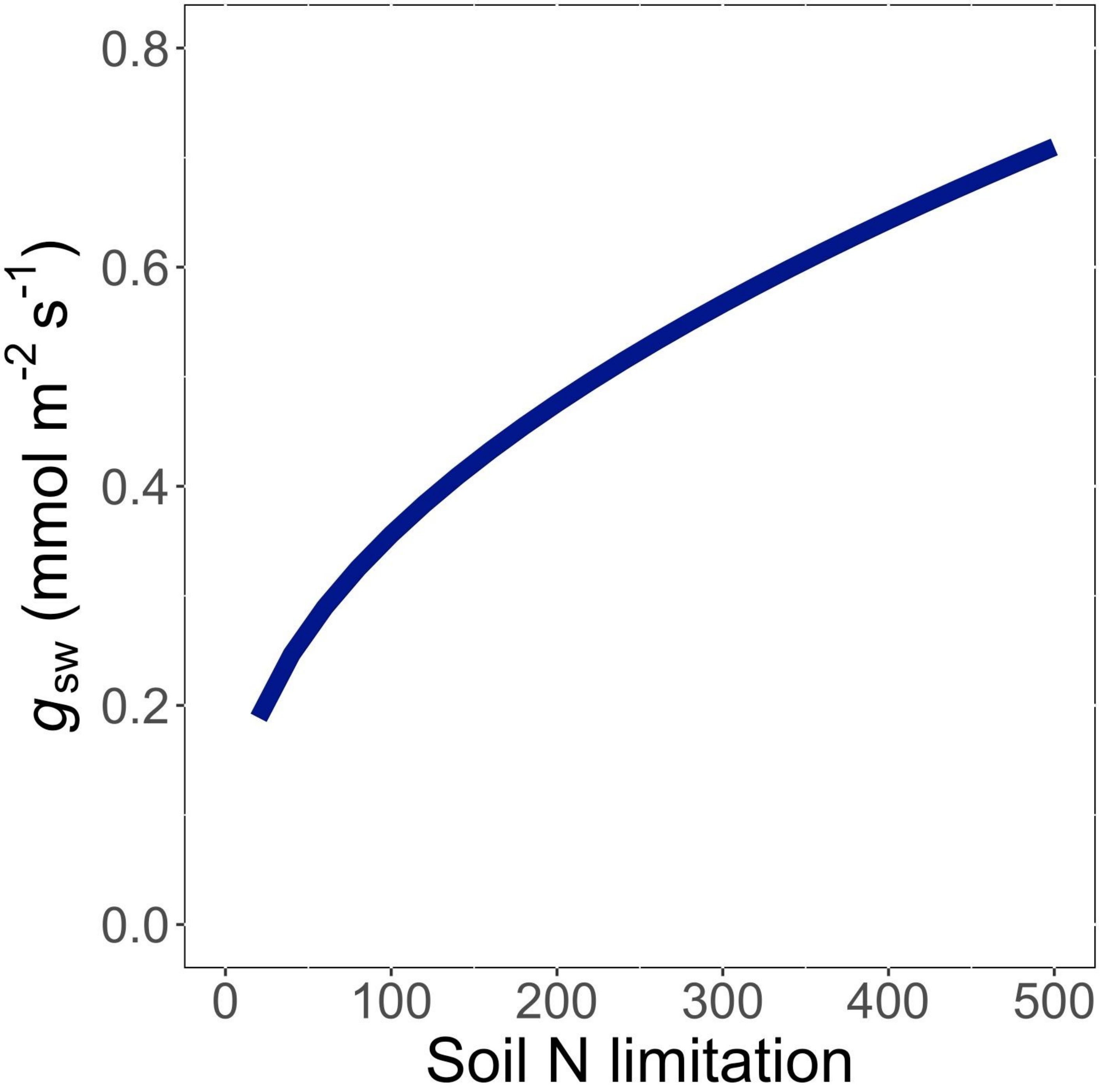
Photosynthetic least cost theory

- Can be used to predict changes in photosynthetic nitrogen in response to changes in
  1. Supply
  2. Demand



Leaf N decreases with soil N limitation, but…

Smith et al. (2019) *Ecol Letters*; Smith and Keenan (2020) *Glob Change Biology*

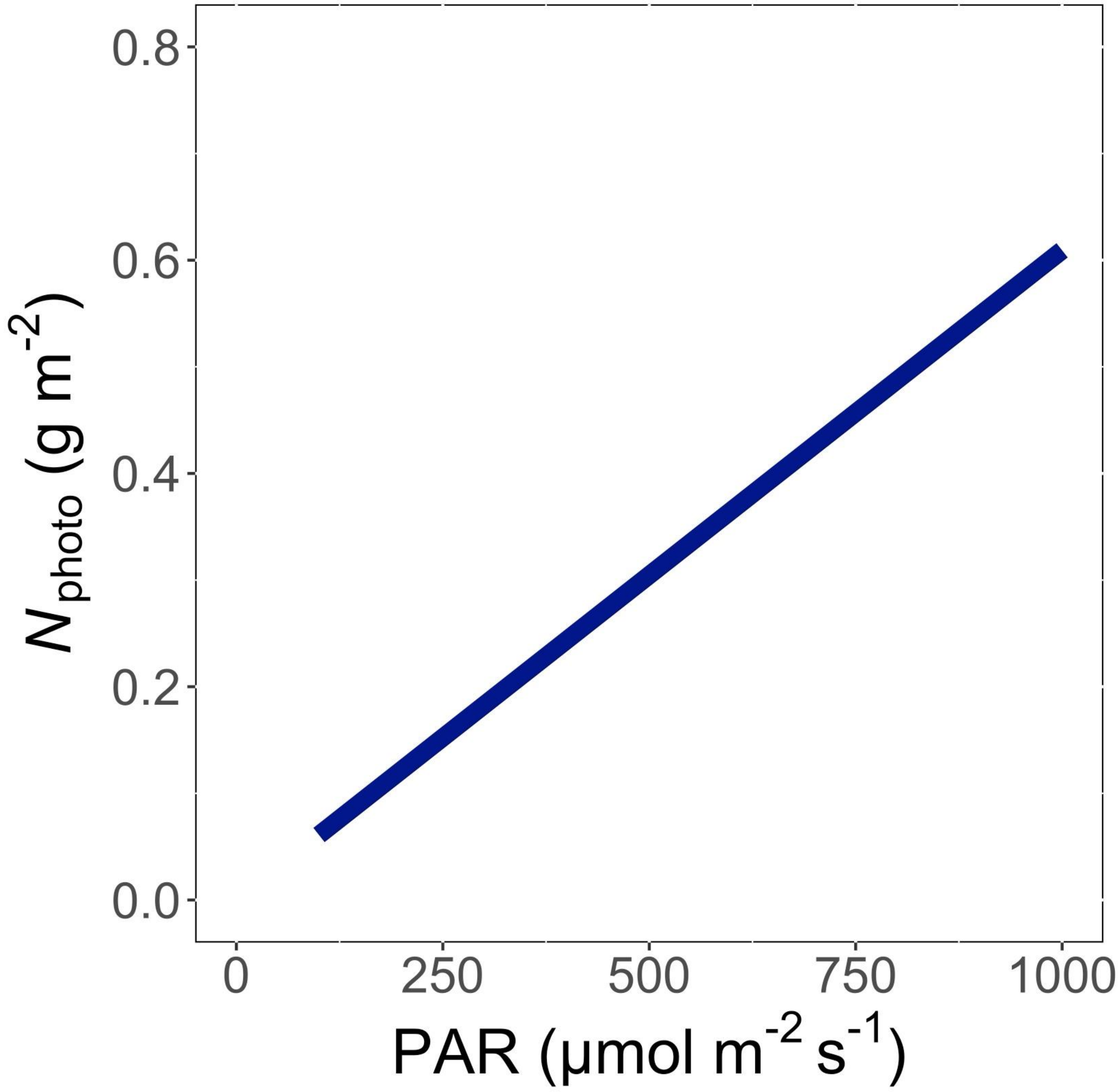


…this requires greater water use (i.e., stomatal conductance) to maintain photosynthesis

Smith et al. (2019) *Ecol Letters*; Smith and Keenan (2020) *Glob Change Biology*

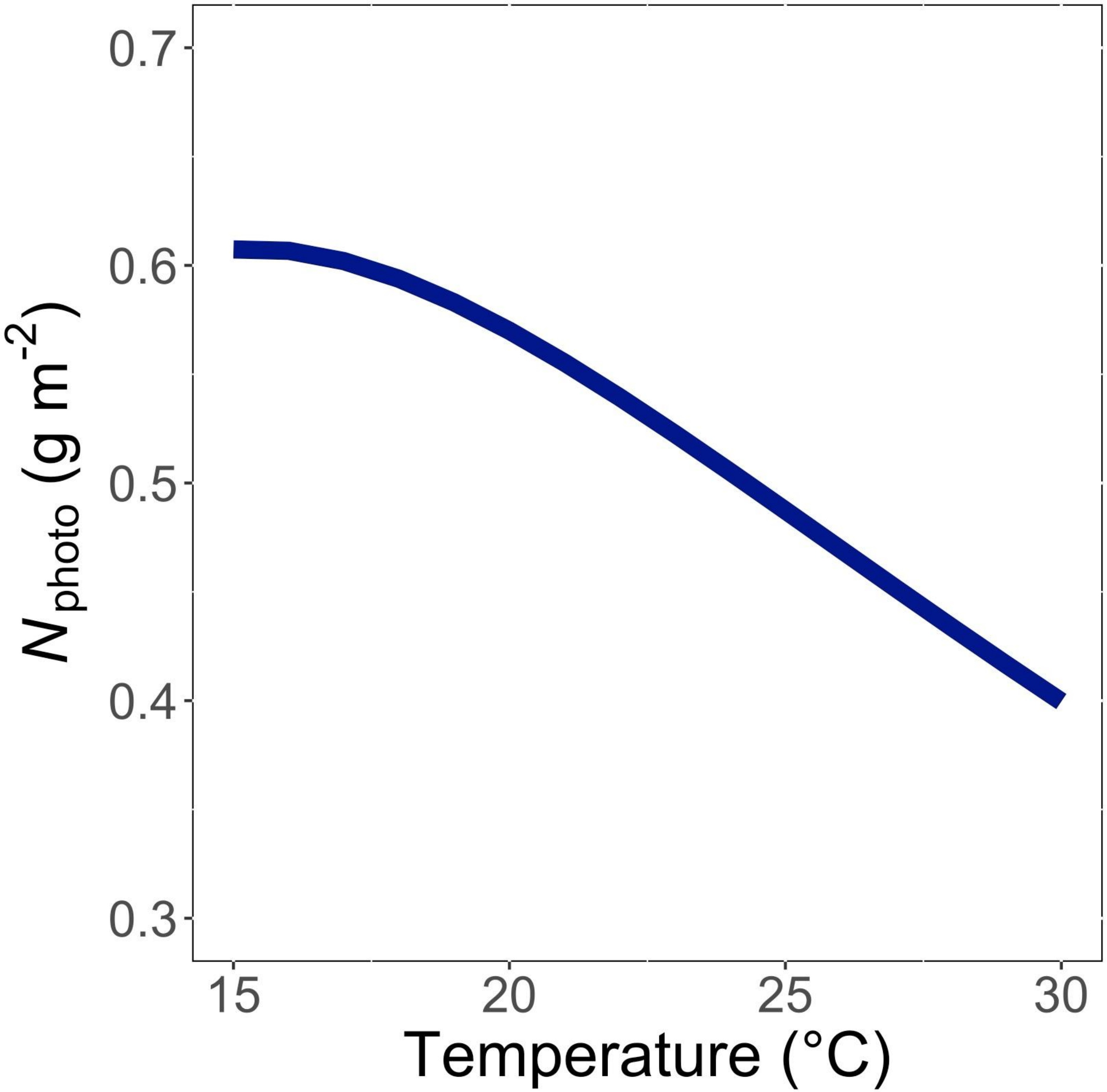
Photosynthetic least cost theory

- Can be used to predict changes in photosynthetic nitrogen in response to changes in
    1. Supply
    2. Demand



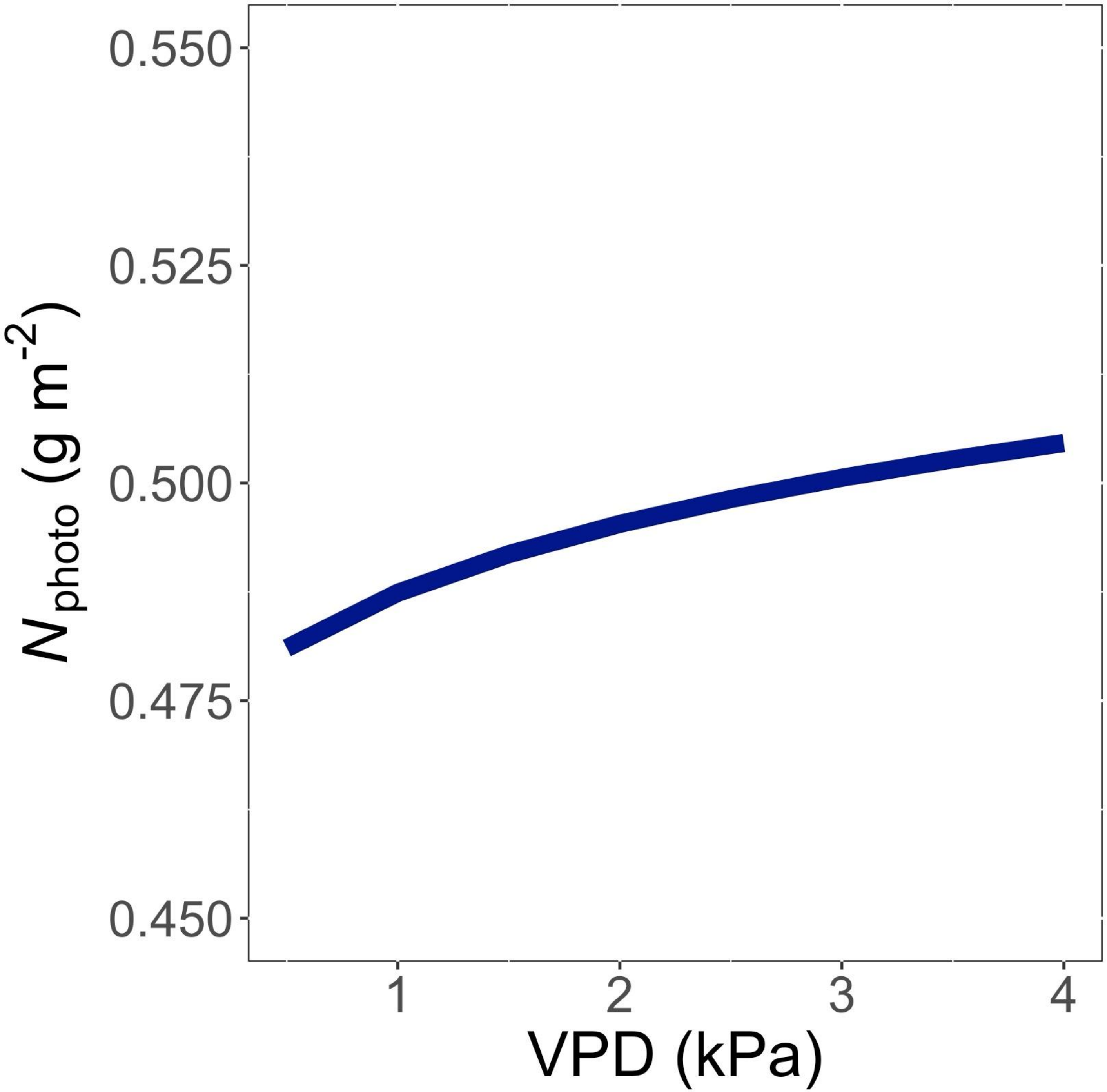
Leaf N demand increases with light because of greater need for photosynthetic enzymes to use light

Smith et al. (2019) *Ecol Letters*; Smith and Keenan (2020) *Glob Change Biology*



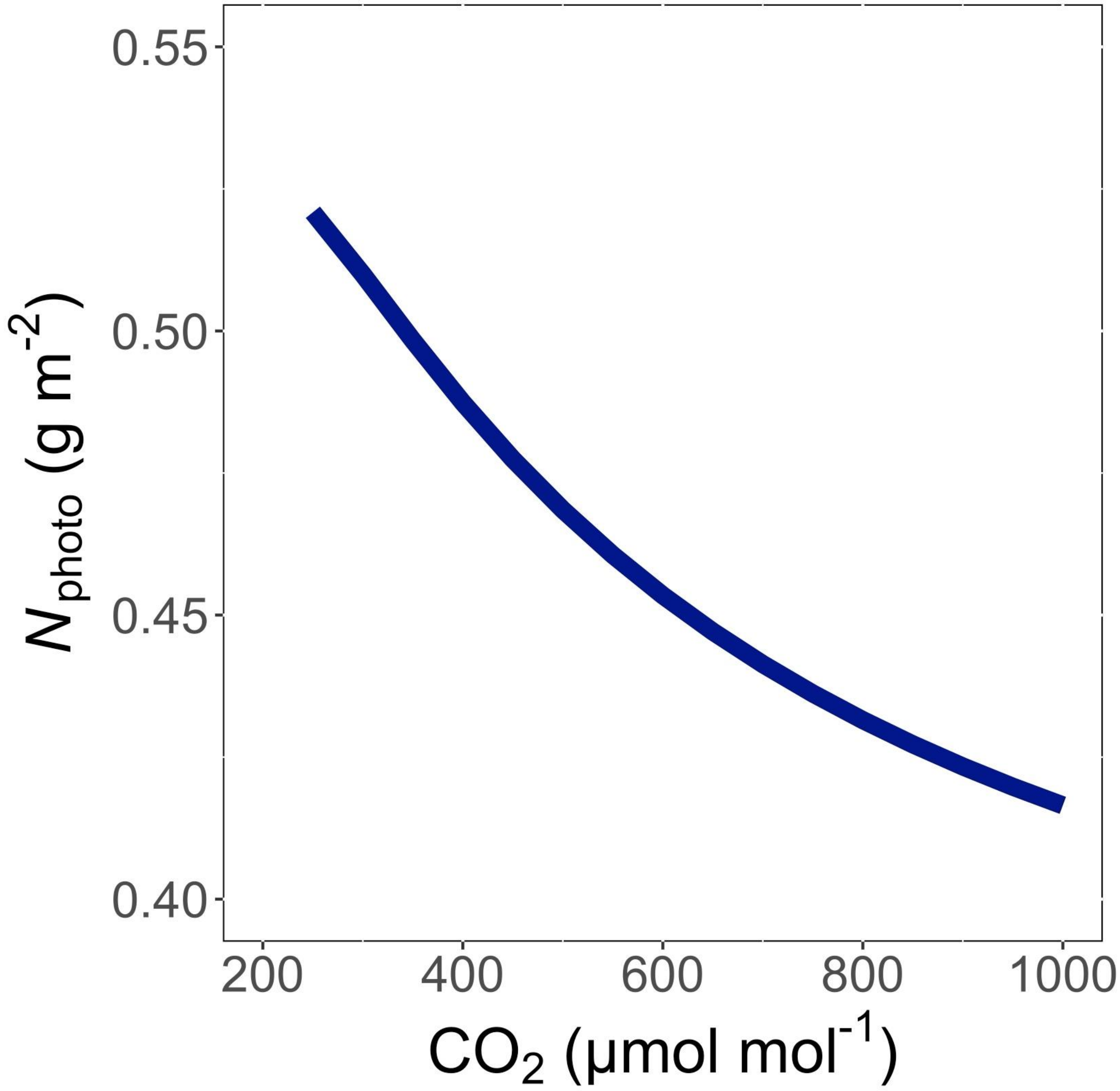
Leaf N demand decreases with temperature because enzymes work faster at higher temperature

Smith et al. (2019) *Ecol Letters*; Smith and Keenan (2020) *Glob Change Biology*



Leaf N demand increases with VPD to allow plants to save water

Smith et al. (2019) *Ecol Letters*; Smith and Keenan (2020) *Glob Change Biology*



Leaf N demand decreases with CO_2 because less enzymes are needed

Smith et al. (2019) *Ecol Letters*; Smith and Keenan (2020) *Glob Change Biology*

Talk outline

- Photosynthetic least cost theory and its relation to nutrients
- Tests of the theory
  1. The interactions between nutrient supply and demand
  2. Responses to elevated CO_2
- Conclusions

Journal of Experimental Botany

Volume 72, Issue 15
28 July 2021

Article Contents

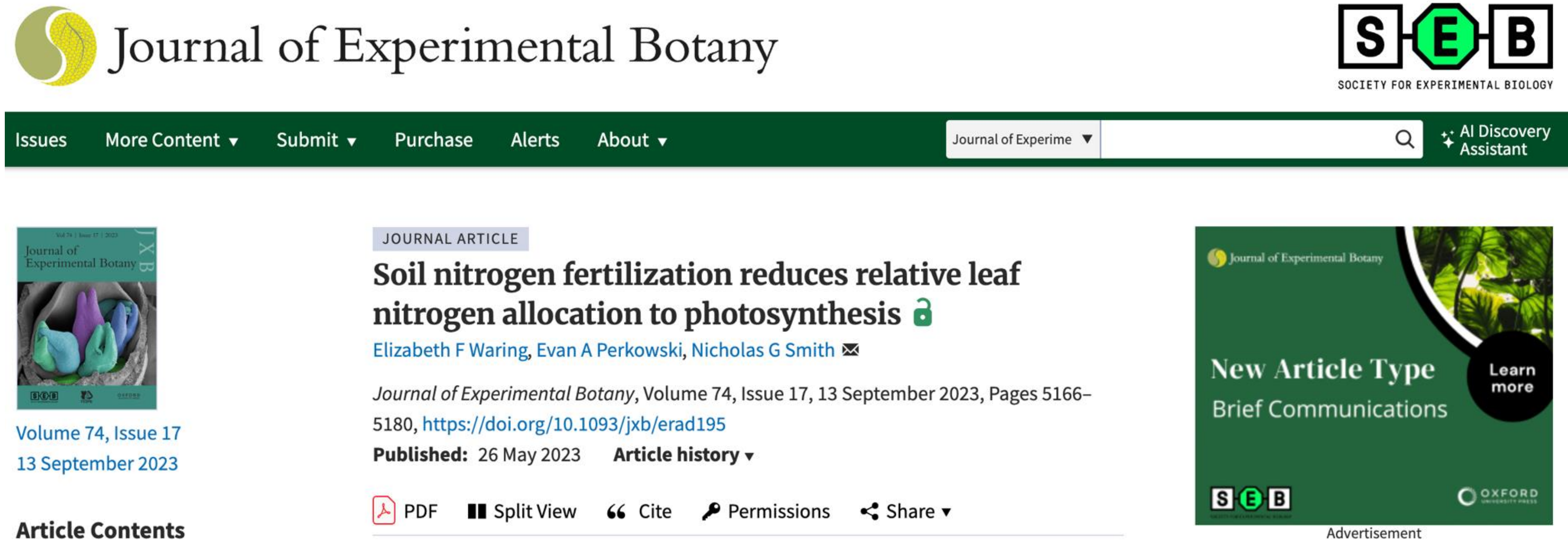
JOURNAL ARTICLE

Root mass carbon costs to acquire nitrogen are determined by nitrogen and light availability in two species with different nitrogen acquisition strategies

Evan A Perkowski ✉, Elizabeth F Waring, Nicholas G Smith

Journal of Experimental Botany, Volume 72, Issue 15, 28 July 2021, Pages 5766–5776, https://doi.org/10.1093/jxb/erab253

Published: 06 July 2021 **Article history** ▾



Journal of Experimental Botany

Volume 74, Issue 17
13 September 2023

Article Contents

JOURNAL ARTICLE

Soil nitrogen fertilization reduces relative leaf nitrogen allocation to photosynthesis

Elizabeth F Waring, Evan A Perkowski, Nicholas G Smith ✉

Journal of Experimental Botany, Volume 74, Issue 17, 13 September 2023, Pages 5166–5180, https://doi.org/10.1093/jxb/erad195

Published: 26 May 2023 **Article history** ▾

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Main questions

- How does nitrogen demand and availability jointly control:
  - Carbon costs to acquire nitrogen?
  - Leaf-level physiology?
  - Nitrogen allocation?
- Do these responses differ in non-N fixing (i.e., cotton) and N fixing (i.e., soybean) plants?

Perkowski et al. (2021) *J Exp Bot*

Waring et al. (2023) *J Exp Bot*

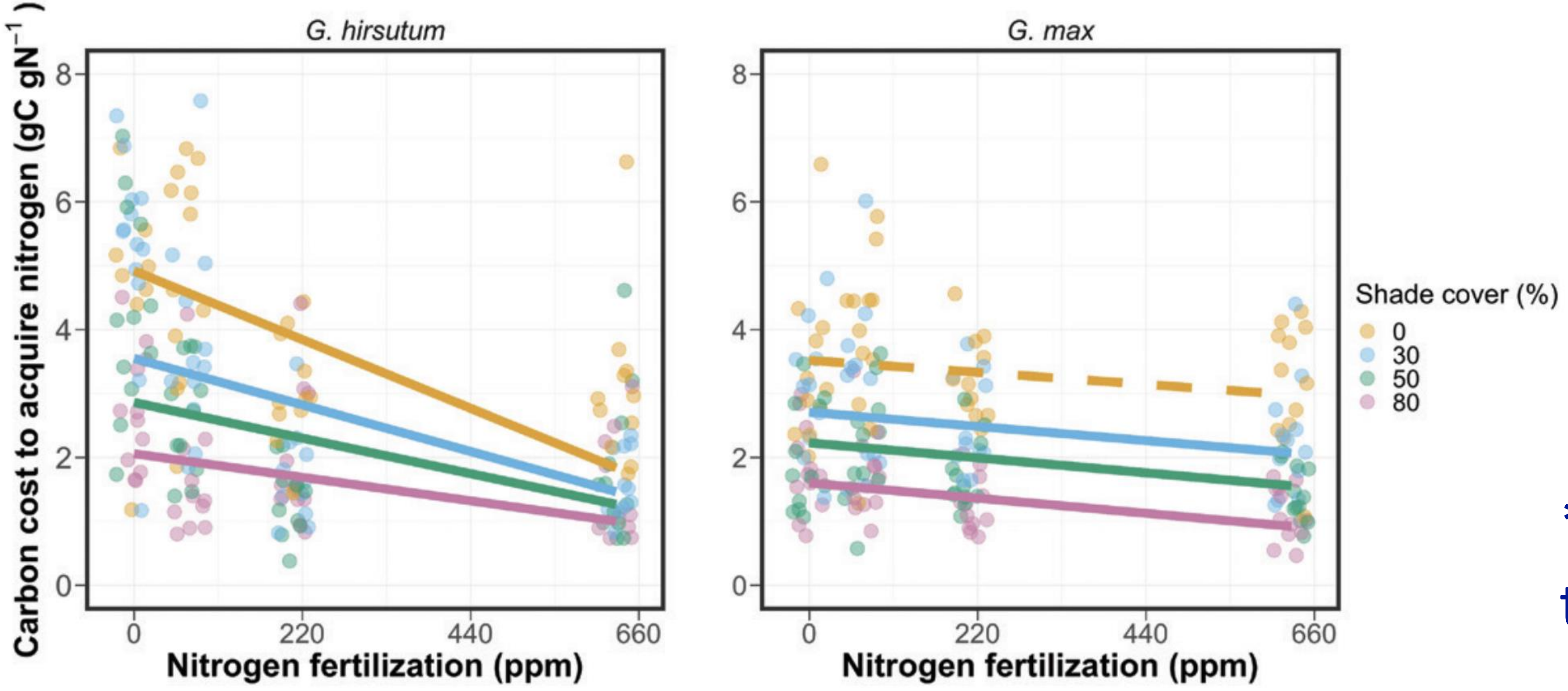
Experimental setup

- Full factorial greenhouse experiment
- Treatments
  - Species: cotton and soybean
  - Light: 4 levels (0%-80% shade)
  - Nitrogen: 4 levels (0-630 ppm Hoagland’s)
- Measurements
  - Biomass and nutrient partitioning among organs
  - Leaf gas exchange and chemistry

Perkowski et al. (2021) *J Exp Bot*

Waring et al. (2023) *J Exp Bot*

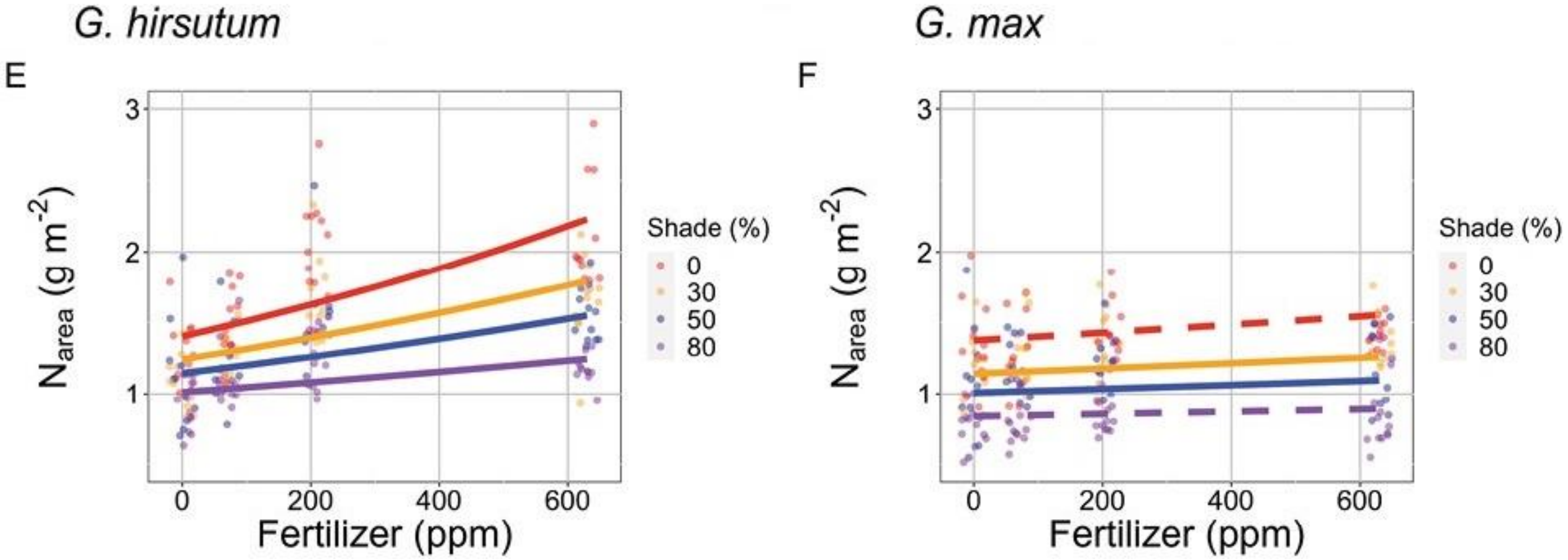
Takehome 1: C cost to acquire N increases with N demand and decreases with N supply*



*Stronger in cotton than soybean

Perkowski et al. (2021) *J Exp Bot*

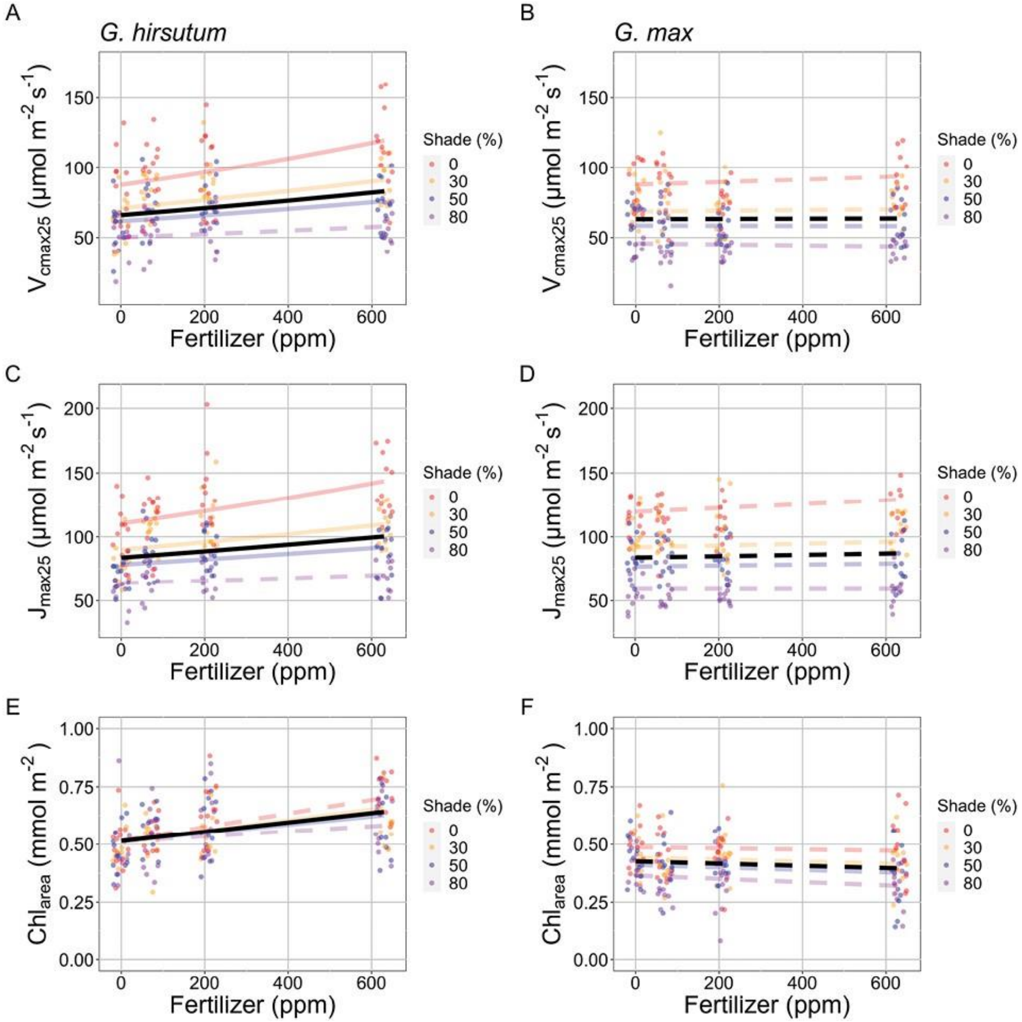
Takehome 2: Leaf N increases with N demand and supply*



*Stronger in cotton than soybean

Waring et al. (2023) *J Exp Bot*

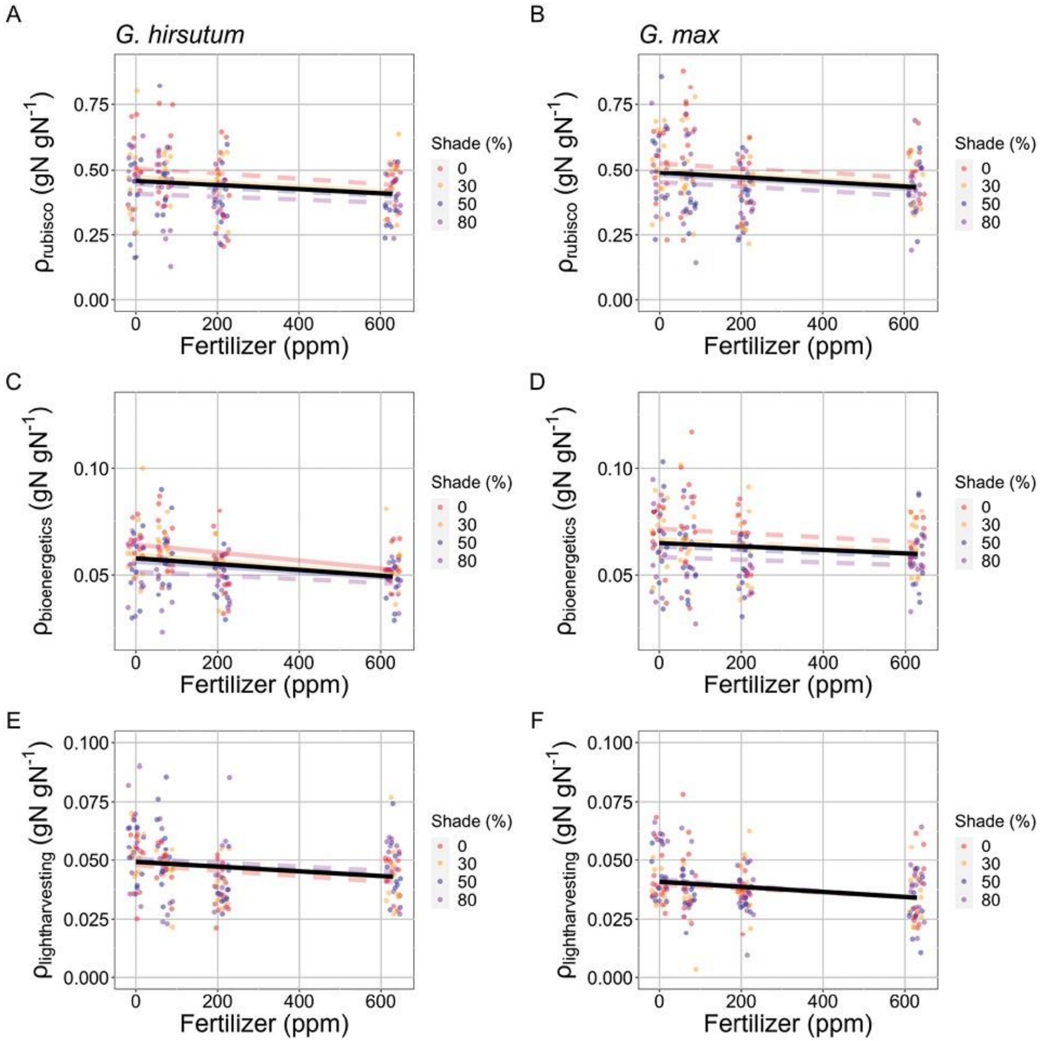
Takehome 3: increased leaf N generally* corresponds to increased photosynthetic capacity



*Stronger in cotton than soybean

Waring et al. (2023) *J Exp Bot*

Takehome 4: most of increased leaf N going to non-photosynthetic processes*



*Similar in cotton than soybean!

Waring et al. (2023) *J Exp Bot*

Nitrogen supply-demand results

- Added soil N makes it cheaper for plants to obtain N regardless of demand
    - Relative impact greatest when demand is greatest
- Both supply and demand impact leaf N
- Leaf N changes with N demand correspond to photosynthesis
- Leaf N changes with N supply do not correspond to photosynthesis

Perkowski et al. (2021) *J Exp Bot*

Waring et al. (2023) *J Exp Bot*

Nitrogen supply-demand results

- Added soil N makes it cheaper for plants to obtain N regardless of demand
    - Relative impact greatest when demand is greatest
- Both supply and demand impact leaf N
- Leaf N changes with N demand correspond to photosynthesis
- Leaf N changes with N supply do not correspond to photosynthesis
- All responses stronger in cotton than soybean, confirmed to be due to N fixation (Perkowski et al., 2024)

Volume 16, Issue 5
October 2024

JOURNAL ARTICLE

Symbiotic nitrogen fixation reduces belowground biomass carbon costs of nitrogen acquisition under low, but not high, nitrogen availability

Evan A Perkowski, Joseph Terrones, Hannah L German, Nicholas G Smith

AoB PLANTS, Volume 16, Issue 5, October 2024, plae051,
https://doi.org/10.1093/aobpla/plae051
Published: 12 September 2024 **Article history** ▾

Talk outline

- Photosynthetic least cost theory and its relation to nutrients
- Tests of the theory
    1. The interactions between nutrient supply and demand
    2. Responses to elevated CO_2
- Conclusions

Journal of Experimental Botany

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AI Discovery Assistant

Volume 76, Issue 10
2 July 2025

JOURNAL ARTICLE

Nitrogen demand, availability, and acquisition strategy control plant responses to elevated CO_2

Evan A Perkowski ✉, Ezinwanne Ezekannagha, Nicholas G Smith

Journal of Experimental Botany, Volume 76, Issue 10, 2 July 2025, Pages 2908–2923,
https://doi.org/10.1093/jxb/eraf118
Published: 17 March 2025 **Article history** ▾

Perkowski et al. (2025) *J Exp Bot*

Main questions

- How does nitrogen supply influence photosynthetic demand-driven responses to changes in in CO_2?

Perkowski et al. (2025) *J Exp Bot*

Main questions

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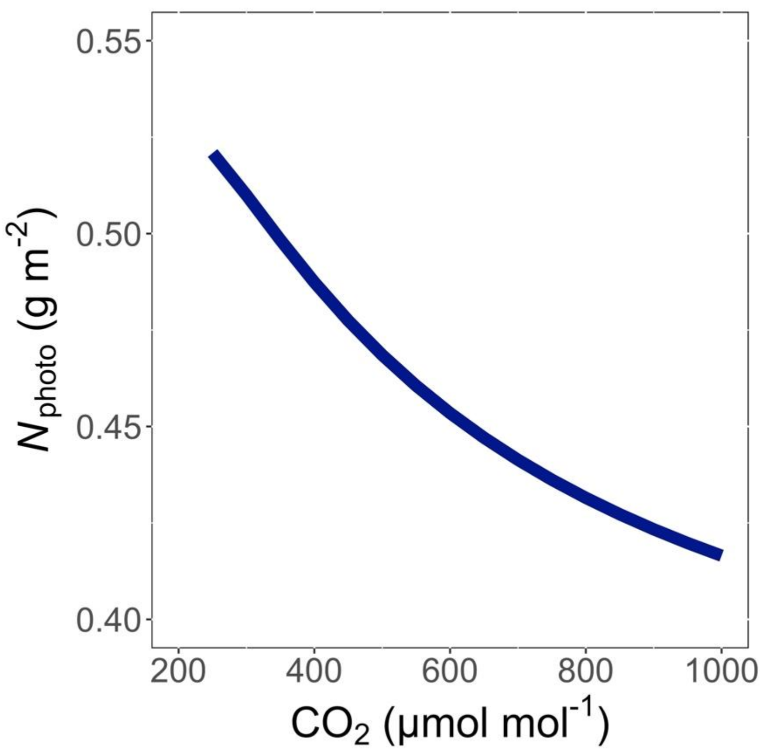
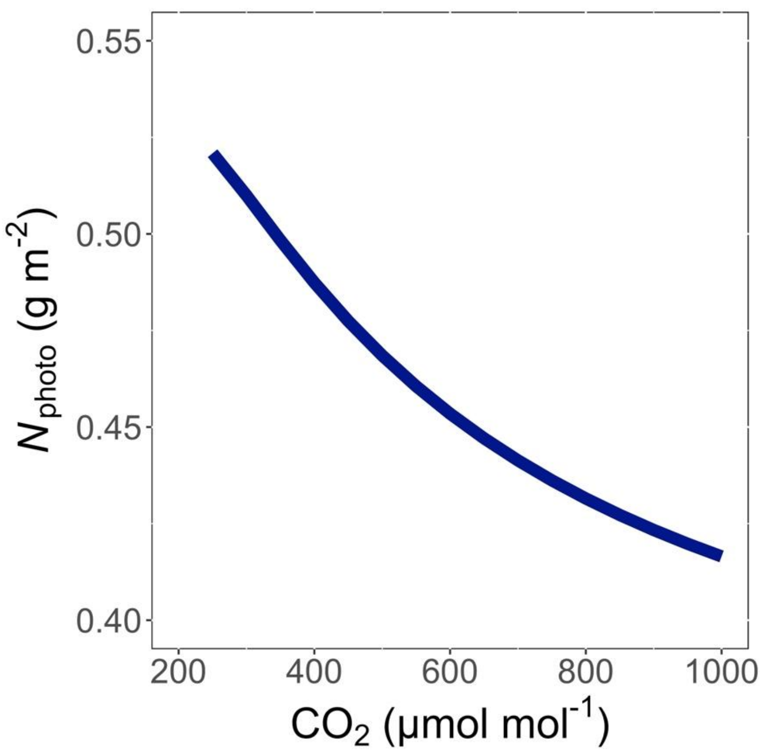


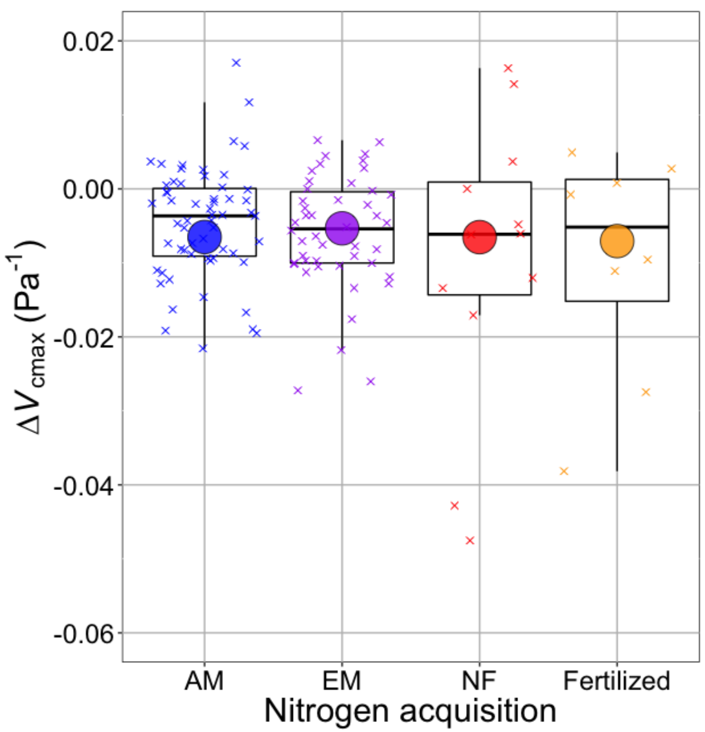
Smith and Keenan (2020) *Glob Change Biol*

Perkowski et al. (2025) *J Exp Bot*

Main questions

- How does nitrogen supply influence photosynthetic demand-driven responses to changes in in CO_2?





Boxes = data = -0.0063 Pa^{-1}
Circles = predicted = -0.0066 Pa^{-1}

Smith and Keenan (2020) *Glob Change Biol*

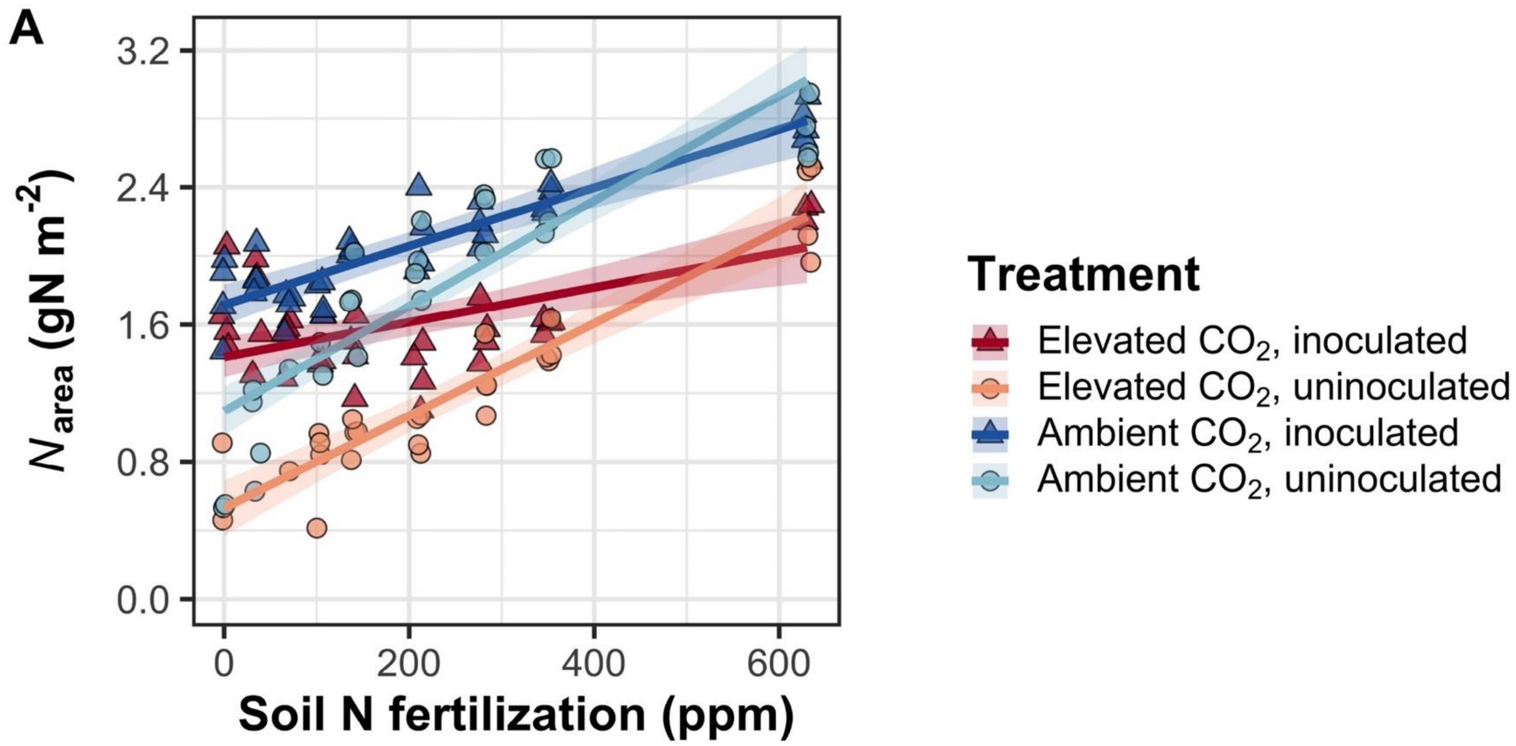
Perkowski et al. (2025) *J Exp Bot*

Experimental setup

- Full factorial growth chamber experiment
- Treatments
  - Species: soybean w/ and w/o rhizobia
  - CO_2: 2 levels (400 & 1000 ppm)
  - Nitrogen: 0 levels (0-630 ppm Hoagland's)
- Measurements
  - Biomass and nutrient partitioning among organs
  - Leaf gas exchange and chemistry

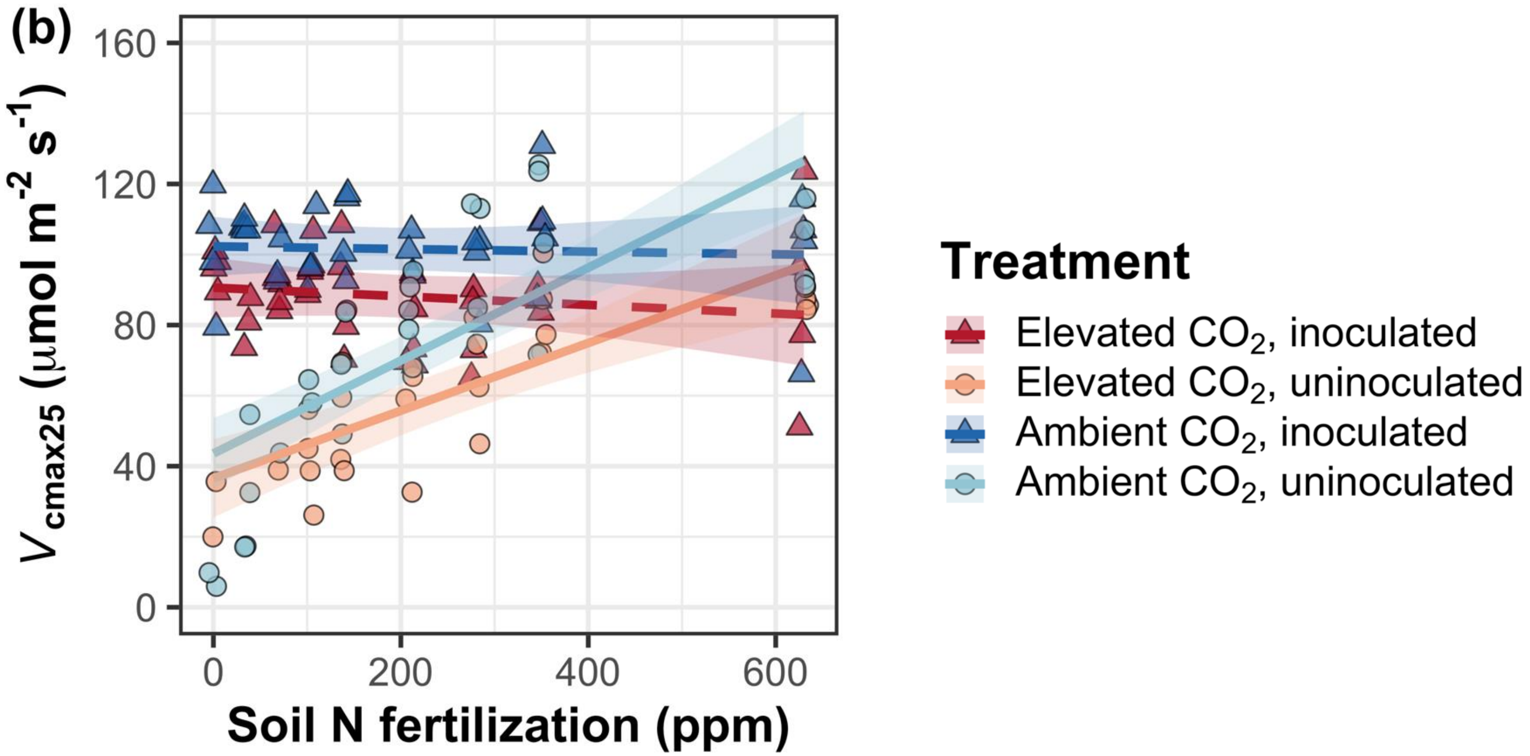
Perkowski et al. (2025) *J Exp Bot*

Takehome 1: N supply increases leaf N, CO_2 decreases leaf N



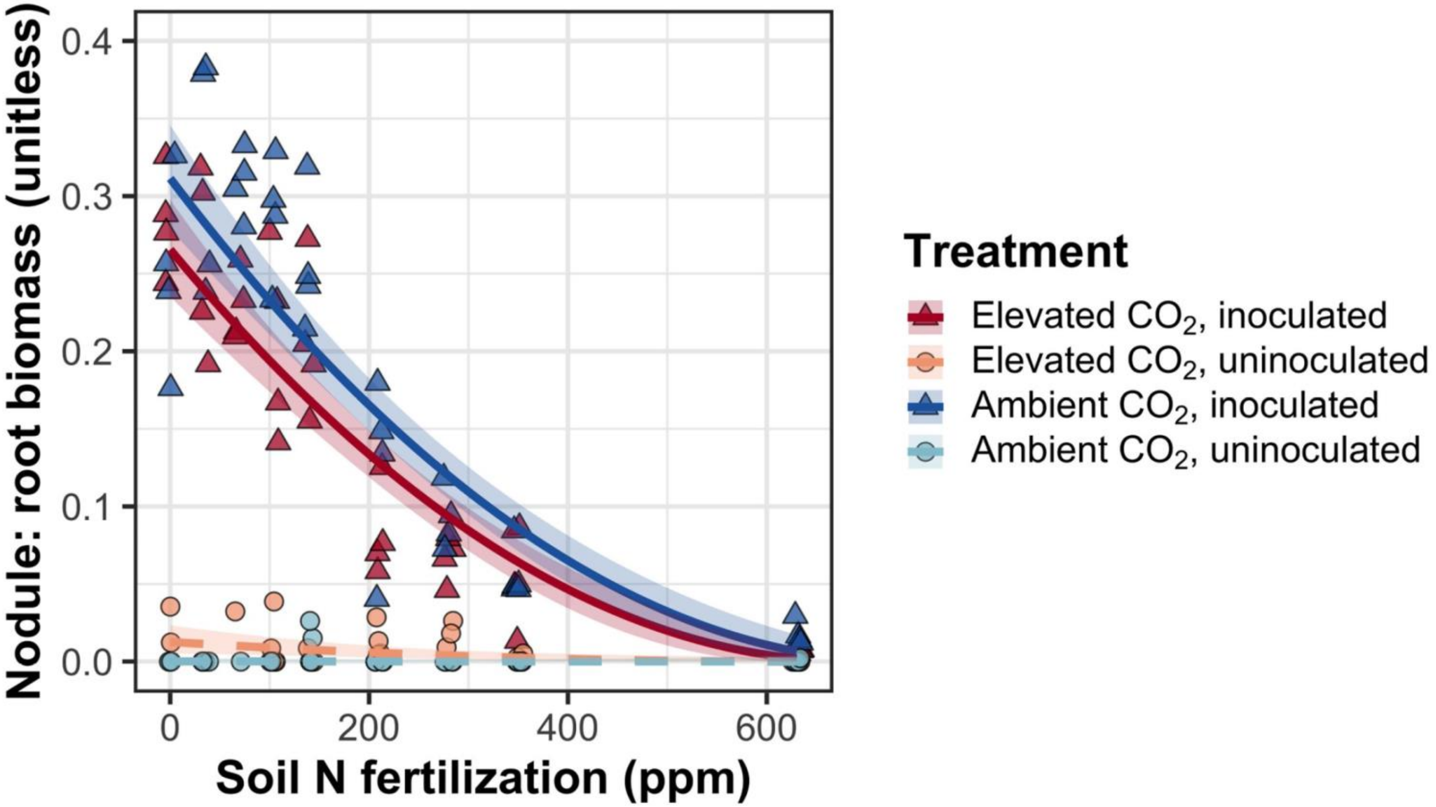
Perkowski et al. (2025) *J Exp Bot*

Takehome 2: CO_2 effect is consistent and due to reduction in photosynthetic capacity



Perkowski et al. (2025) *J Exp Bot*

Takehome 3: reliance on N-fixing bacteria decreases with increasing soil N



Perkowski et al. (2025) *J Exp Bot*

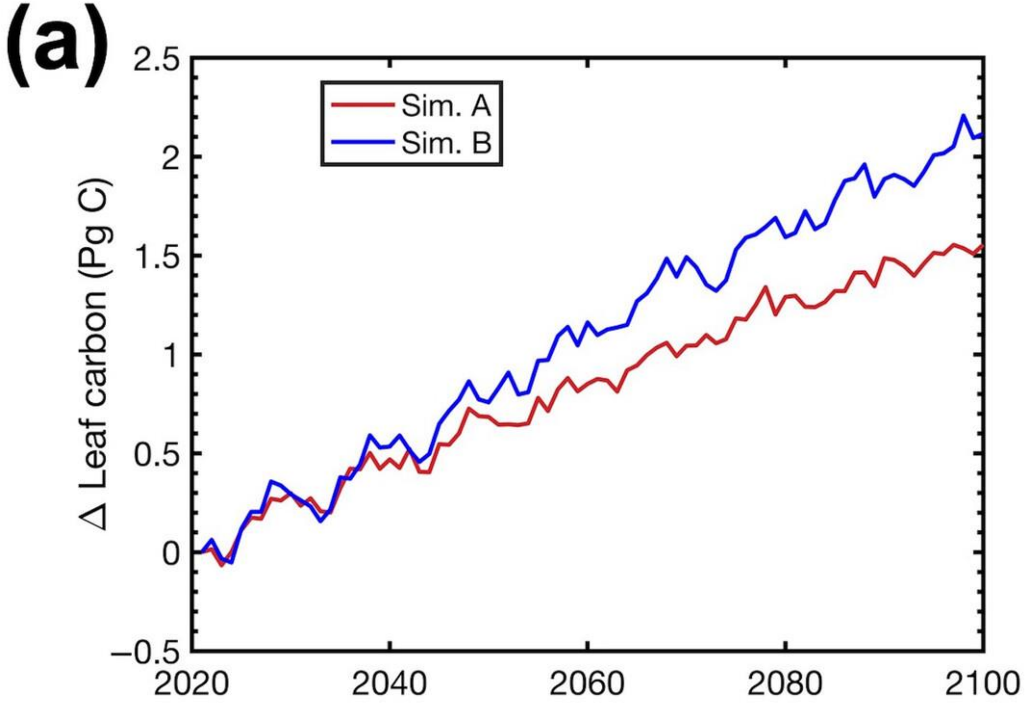
CO_2-N results

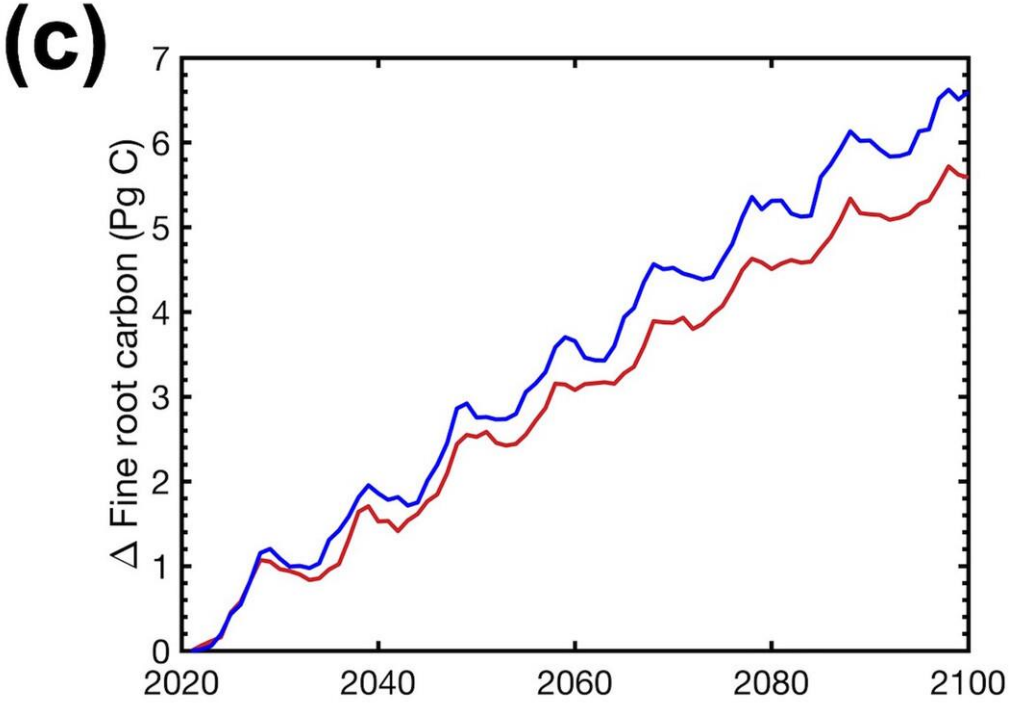
- Little interaction between soil N supply and CO_2-driven changes in N demand
  - Soil N supply increases leaf N and photosynthetic capacity
  - CO_2 reduces photosynthetic demand, leaf N, and photosynthetic capacity (independent of soil N)
- As in previous experiment, soil N supply responses are reduced in individuals capable of N fixation

Perkowski et al. (2025) *J Exp Bot*

What do plants do with this saved N?

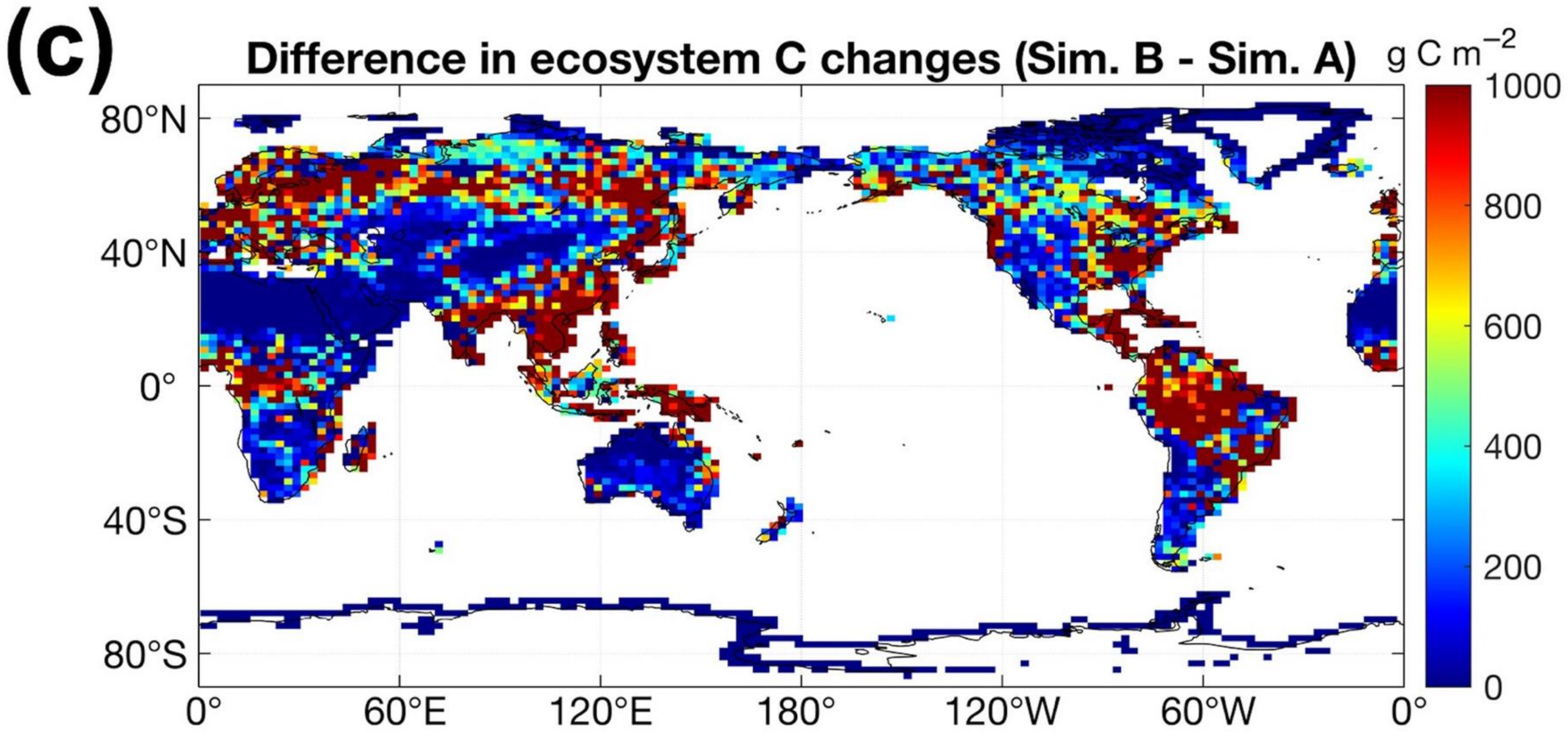
When plants save N under elevated CO_2, they build more leaves and roots





Smith et al. (2024) *Glob Change Biol*

Leaf N savings increases long-term ecosystem carbon stocks



50% increase in ecosystem carbon from CO_2 acclimation

Smith et al. (2024) *Glob Change Biol*

Talk outline

- Photosynthetic least cost theory and its relation to nutrients
- Tests of the theory
    1. The interactions between nutrient supply and demand
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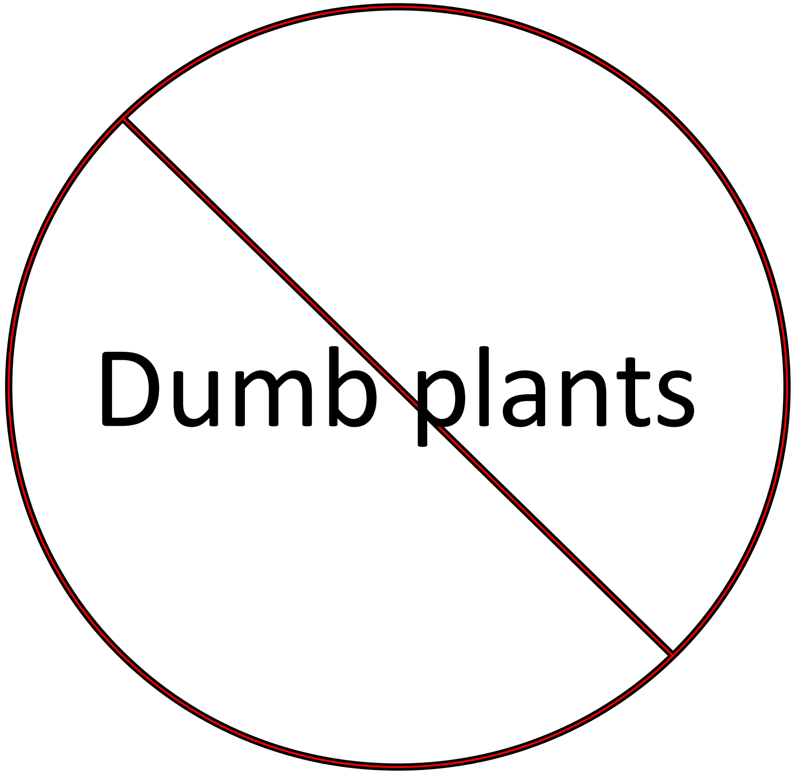
Conclusions

- Plants aren't dumb!
  - They respond dynamically to changes in their environment
  - This includes changes in nutrient availability and nutrient demand



Conclusions

- Quantified theory can help to better understand the mechanisms underlying these response dynamics
  - Can be used to predict plant-to-ecosystem level responses to environmental variability

Presentation available at:
www.github.com/SmithEcophysLab/seminar/gpsfc_2026

Contact: nick.smith@ttu.edu